Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/28/2014
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Project Title:	Electric Field Control of Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-3975	**Congressional District:**	48
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2011	**End Date:**	09/30/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2016 per NSSC and D. Stocker/GRC (Ed., 9/24/14)		
Key Personnel Changes/Previous PI:	August 2013 report: Professor Felix Weinberg passed away in December of 2012.		
COI Name (Institution):			
Grant/Contract No.:	NNX11AP42A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE this is a successor agreement to "Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)," per D. Stocker, NASA Glenn Research Center. This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving the performance of energy conversion systems. As an actuator, electric fields can affect flame stability and soot formation; while as a sensing device, the electrical response at saturation is a flame characteristic. In addition to preparing for the proposed ACME E-FIELD experiments aboard the ISS, we are currently studying electric field effects on quenched jet flames, partially-admixed air/fuel jet flames, and their associated dangers. This report summarizes recent findings. Specific activities include: 1.) Study electric field control of flames with wall effects near quenching. The experiment involves flow field imaging,		

Task Description:	along with OH chemiluminescence and OH PLIF techniques. Measurements include thermal emission and carbon monoxide emission, and analyses evaluate thermal emission. 2.) Characterize time-scales related to the interaction between ions and neutral molecules within the burner/mesh electrode gap. These time-scales provide information into how an electric field affects combustion related parameters. These times-scales were determined using high-speed schlieren imaging and ion current/applied voltage measurements. Our understanding developed in these areas may improve kinetic mechanisms for diffusion flame modeling (including charged species), help clearly identify the risk of CO emission from quenched flames, and help determine if electric fields can actively control the release of carbon monoxide from flames. Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways. First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way. Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The control of combustion has the potential to improve efficiency and reduce emissions from burning fuels. Since high power density often requires combustion, these improvements will be important no matter what the fuel source. Electric fields acting on flames have the potential to aid in combustion control. In addition, electrical properties of flames can identify poor performing boiler flames that release poisonous carbon monoxide. Our studies show that a flame's electric signature can capture incipient quenching before dangerous emissions result. Understanding the links between electrical character and flame behavior may allow improved sensing of poor performing combustion systems.
	UCI Coflow Burner--All ground-based experiments were performed using a coflowing burner that is dimensionally similar to the proposed ISS coflow burner. The burner consists of two stainless steel plenums threaded together. Fuel, introduced in the bottom plenum, is passed through a stainless steel tube (2.13 mm ID) and exits the top surface fully developed. The air entering the top chamber passes through flow straightening beads and a honeycomb mesh. The air exits the top surface with a top hat flow profile at the exit. A variable high-tension power supply connected between the burner and downstream mesh electrode produces the applied electric field. The electric field distribution is deformed by the flame’s space charge, but defining an applied field strength, E = V/d, is a useful approximation when describing experimental results. Ion current is computed using Ohm’s law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber shields the flame from room currents. Quenching studies--The effects of quenching on flame structure study was initiated by Professor Felix Weinberg at Imperial College in order to predict the hazardous conditions created by poorly tuned industrial boilers. The ultimate practical aim of this series of experiments is to use electrical aspects of flames to reduce the incidence of CO poisoning caused by the quenching of natural gas flames in malfunctioning burners. We investigate, for example, how the flow of charge can be employed to signal the onset of hazardous conditions before any CO release can occur. This research has now shifted entirely to UC Irvine as Professor Felix Weinberg passed away last year. The work involves the examination of the electrical properties of partially quenched flames. Our work at Imperial College illuminated the potential dangers of improperly tuned boilers and the high concentration of carbon monoxide and hydrogen potentially emitted when quenched. This portion of our research was aimed at understanding the electrical behavior of a flame as a plate was incrementally lowered toward it by measuring the corresponding ion current. The study was a proof of concept to explore the potential of detecting flame impingement electrically without visual inspection. The work has shown that ion current sharply increases and finally saturates when lowering the plate toward the jet flame exit. The ion current decreases as the flame temperature and ion production fall. We continued this work to study the phenomena occurring on the quenching plate as the flame approached it. Different

quenching surface materials have various emissivities and thermal properties that lead to different equilibrium temperatures. Preliminary research involves identifying the thermal emission from the plate at different heights when lowered towards the flame (or when the flame sheet is interrupted).

A z-type Schlieren imaging technique is used to understand how the flow field changes with electric field control. The results of CO emission and flame imaging show that the electric control of flames has a clear effect when H/D (H – distance to the plate; D – fuel jet diameter) values are larger than 4.5, while the cases with smaller H/D create complex flow regions because they are dominated by the jet momentum. Schlieren images with and without electric field applied to the flame show steady laminar flow fields, with a typical stagnation point flow boundary layer shape extending radially outward on the plate to its edge.

An infrared camera (FLIR SC620) measures radiative emission and is situated directly above the quenching plate at normal incidence to avoid complications associated with Lambert's cosine law. The quenching plate is painted with black paint and is considered a blackbody object. This assumption has been confirmed as reasonable using thermocouple temperature comparisons at a few spot locations. By using the heat map to characterize the spatial concentration of the heat balance on the plate helps describe how electric fields can control the concentration of the heat. Thermal maps from the 2-D image are converted into a radial temperature profile across the plate for comparison at H/D 4.5 (9 mm). The overall temperature distribution is approximately Gaussian. With negative field, the peak temperature is lower and profile is broader while the positive field produces higher peak temperature and narrower profile. This information suggests strongly that the heat transfer over the quenching plate can be altered or improved by an electric field and in so doing may provide much better thermal heating efficiency, particularly if the target plate is small.

Previous results convey the information that the total heat transfer is not affected substantially by the electric field near quenching at cases below H/D at 4.5 (9 mm); so, another assumption needs to be applied for these cases. The temperature profile in thermal maps indicates that the temperature profile does change with electric field. Therefore we assume that the field affects the local heat flux distribution rather than the total flux to the plate when the gap distance is small. The concentration of heat over the plate can be analyzed in the following way; there are six rings mapped from the hottest spot, these six rings are different percentages of the total heat loss (and thereby heat gain at steady state) from the plate. We next study how the heat flux changes with electric field strength. The heat flux is growing as the rings moves close toward the hot center. The heat flux variation under electric field effects at 80% to 50% is around 400 to 700 W/m2, the 40% ring is 800-900 W/m2, the 30% ring is 1000-1100 W/m2, and at 20% the flux can be up to 1400 W/m2 change in the flux when an electric field is present. These results show conclusively that the electric field can change the local heating from a flame in a significant way, particularly near the center cone, as expected.

A close look of the 20% heat loss with normalized heat flux at zero-field at each height shows the effect of plate location. The electric field seems to have the strongest effects at the larger distance of H/D = 5.5. This seems to occur because the tradeoff between directing the heat and quenching the flame begins to occur at approximately this height. The H/D 1.5 (3 mm) case is very complex since the plate is very low and the plate impeded flow field interacts with the flame in some way. We need to examine these extreme cases in more detail. The temperature and heat transfer over a quenching plate with the effect from electric field flames is investigated from H/D 5.5 to 1.5 (plate location 11 mm progressively lower down to 3 mm) with the maximum electric field that can be applied without breakdown. The plate has a broadened profile and lower peak temperature at the negative field; with the increasing of the field to the positive the plate has higher peak temperature. The overall heat flux to the plate before quenching increases with the field. Under conditions where the flame is being quenched by the plate, it is interesting to note that the variation with the electric field decreases. For example, at the 20% ring, the heat flux is increasing from 2400 W/m2 up to 3850 W/m2 at H/D 5.5 (11 mm) as the field focuses the flame. Similar phenomenon can be observed during flame quenching but these changes are smaller as compared to the cases without any flame/plate interaction. Further exploration is required such as on how does the heat propagate and distribute at the bottom side of the plate.

An initial measurement to understand the overall carbon monoxide emission inside a chamber shows how the combustion reaction alters under electric field control. Reactions consuming hydroxyl compete with each other layer by layer within methane/air flames, and carbon monoxide often employs hydroxyl to oxidize. OH planar laser induced fluorescence (PLIF) shows that OH location and concentration during flame quenching correlate strongly with carbon monoxide release. The PLIF measurements follow generally recommended practice and are conducted by pumping OH at 282.0 nm in the A-X(1,0) band and detecting fluorescence in the A-X(0,0), (1,1) band. The system is composed of a Nd:YAG laser (YAG) and a pulsed dye laser (PDL) with a wavelength extender (WEX). Being a legacy laser system, it is not automated and the experiment requires manual tweaking of the crystals to find and maintain good performance. The requirement for continuous adjustment makes precise quantification difficult but, as shown by the results, the manual effort produces very clear evidence of OH distributions in the impinging diffusion flames. The pulse energy was kept below 1 mJ to stay in the linear regime of the fluorescence. A high speed Phantom camera (v4.3) and a HiCaTT intensifier equipped with a P24 phosphor collected the images. A band pass filter centered at 320 nm was used (Semrock FF01-320/40-25) which offers 70% transmissivity to collect the light from the A-X(0,0) transition. The main peaks that are found and suitable for PLIF imaging are the stronger ones that start at 282.0 with the Q1(1) transition followed by the Q1(2), Q1(3). Though it is more sensitive to temperature changes than the most used transition Q1(8), the combination of lines still makes a very good choice for 2D temperature PLIF measurements if used in combination with other very sensitive temperature transitions. In addition, this transition range allowed less intensifying, leading to a better signal to noise ratio, and for the purposes of the current study, the 10% variation is not significant relative to the natural variations that occur in the electrically-driven flames being measured.

OH* natural chemiluminescence images in the band from 300 nm to 340 nm (using the same filter as before) are also recorded at gating around 10 microseconds. The images are processed through an Abel transform to reveal tomographically reconstructed local distributions. The reconstructed images are after averaging 200 pictures taken at a rate of 100Hz and after filtering with its background. The transformed images show a clear region of the reaction zone. In general, the overall OH intensity decreases as the plate approaches the burner, which implies that OH radical concentration decreases at smaller quenching distance. This, combined with the fact that the velocities were kept low so the flow behaves with similar characteristics of a laminar stagnation flow, gives a reason for the overall oxidation of the CO molecule to decrease, due to the lower OH concentration and less residence time for reaction through the OH layer as the plate approaches the burner. The tip of the OH radical zone opens because the plate acts a chiller for the flame, radiating energy and bringing the surface temperature to approximately 500K. Other work mentions that the OH radical is not expected to exist below 1000K, which is why the tip of the OH reaction zone opens in the cool gas layer near the surface even before the HR zone changes its shape. In addition, the overall OH fluorescence changes with electric field and distributes outwardly with negative fields while it gathers with a positive field. At relatively high burner to plate

Task Progress:

spacing the electric field has minor influence on the OH PLIF intensity but it does affect the location of the OH as the field affects the location of the flame sheet. When the electric field is applied upward, the ion driven wind pulls the flame in the direction of the flow field and the reaction became vigorous toward the plate. OH distributes strongly near the plate in this case. At closer plate distances the field affects the OH intensity as well as its position. When the electric field is negative, the downward directed positive ions produce a downward blowing wind that causes the reaction sheet to widen. When H/D at 3.5 (7 mm), there is the most dramatic effect of the electric field as regards the relative magnitude of OH fluorescence but relatively little change in the location of the OH relative to the flame sheet. This result is reflected in the large relative change in CO for the case with negative fields applied. As with the CO measurements, these images show that the effects of changes in plate height have a larger effect on the OH distribution than does the electric field.

Chemi-Ion Driven Flows in a Time-Varying Electric Field--For similar current densities, a flux of H3O+ ions through a gas, such as air, will produce body forced per unit volume about three orders of magnitude greater than a flux of electrons. In order to test this theory an experiment was needed to measure the velocity field within the electrode space region. A simple schlieren system is employed to visualization of the thermal plume within the electrode space. In the absence of an electric field, the thermal plume is buoyantly driven due to the pressure forces of the cool ambient gas environment acting on the hot combustion gases. The buoyancy force acting on the bulk gas can be described by the Boussinesq approximation. The upward motion of hot exhaust gas has been known to entrain ambient gas horizontally from the environment in order to satisfy mass conservation. With the presence of an electric field, the thermal plume is driven by both buoyancy and electrical body forces. An electrically driven thermal plume has been shown to entrain ambient gas similar to buoyancy driven thermal plumes but is expected to be higher by an order of magnitude for low heat release rates (i.e. low fuel flow rates). Schlieren technique is providing insight into time-scales related to the electro-hydrodynamic parameters that are being investigated in this study.

A brief theoretical analysis of the time-scales is performed before proceeding to the results of the schlieren experiment. The first time-scale is the time it takes to pull a charged particle from the flame and retrieve it downstream at the mesh electrode. For a positive field, H3O+ is the primary ion thought to exist within the electrode gap. The velocity of a single H3O+ ion travels through the electrode space at its drift velocity is equal to the product of its mobility, K, and the field strength, E. The time it takes for an H3O+ ion to travel from its place of origin, the flame, to the downstream electrode is then defined by the length of the electric field line divide by KE. Since the flame remains attached to the burner, it is assumed that the burner and flame have the same potential. The charge density only effects the local electric field lines near the flame where the lines are given a small radius of curvature but become parallel lines quickly downstream. This curvature slightly alters the length of the field lines, thus, altering their local magnitude. Since the order of magnitude of the analysis is less important compare to the whole scale of the flow field, the length of the field line will be taken as the length from the base of the flame up to the downstream mesh electrode. This is a good approximation, since it is thought that the highest concentration of H3O+ is located near the base of the flame. The electric field strength will be defined as the applied voltage difference between the two electrodes divided by the length of the electric field line. This analysis uses the applied voltage at saturation to produce the maximum, purely chemi-ionization effect. The length of the electric field line is 3.2cm, K is 2.9cm^3/V and the applied voltage as measured from the downstream electrode with respect to the burner is 5x10^3 V. Then we get the drift velocity 4530 cm/s and the time that the ions travel is 1ms. The next time-scale is the time it takes a volume of gas driven by a current of chemi-ions to travel from the flame to the downstream electrode. For simplification, this analysis will be performed for a one-dimensional system neglecting, viscosity, buoyancy, and pressure gradients within the ion stream. The rate of change of momentum along the axis is equal to the body force acting on the gas integrated over the volume between the flame and the downstream electrode. The maximum velocity due to the ion wind is also calculated as 187cm/s and the average velocity is 124 cm/s. The calculated time a volume of bulk gas to travel through the electrode space is ~34 ms. The analysis neglects buoyancy forces in the bulk gas. For the experiments performed in this study, buoyancy forces cannot be neglected. However, considering buoyancy and ion wind forces together leads to an extremely difficult mathematical problem. Therefore, an analysis of time-scales for buoyancy is performed next for reference. The velocity of a buoyantly driven gas is also approximated by velocity equation, and we get the maximum value 50 cm/s at T=1800K. The average value is 34cm/s. As expected, these values are of the same order but somewhat less than the velocities due to an ion wind. The time for a volume buoyantly driven gas to travel through the electrode space is also calculated around 125ms.

The first test was to test the voltage current response of the electrical/burner system. The voltage vs current of the system with and without the flame curve has characterized. The horizontal axis is the applied voltage on the mesh electrode with respect to the burner and has units of kilovolts. The vertical axis is the measured ion current measured through the shunt resistor and has units of microamperes. The test was for a methane flame with a fuel flow rate of 7cm/s. The co-flow was not used during this test. The negative field shows a saturation current of 1.25 microamps and the positive field shows 1.75 microamps. One last note that should be made is the ion current past the negative saturation point decreases and then increases very quickly. This is most likely due to thermal effects of hot metal surfaces (burner/mesh electrodes) releasing electrons past this high voltage range, i.e. thermal ionization. The applied voltage and ion current versus time in milliseconds are measured during this test. High-speed schlieren video also shows that the thermal plume starts at steady state where the gas velocity is driven only by buoyancy forces in the hot gas. At this state, the plume boundary has ~12mm diameter thickness downstream of the burner. The base of the plume boundary covers most of the extruded fuel tube. After the electric field is switched on, no noticeable movement occurs in the plume boundary till t=10ms. It should be noted that by comparison to ion current versus time curve, the ion current has become fully established just a few milliseconds prior to the 10ms mark. This shows that the interaction between chemi-ions and bulk gas is <10ms. At the instant of first noticeable motion, the base of the plume slightly rises and the diameter of the plume decreases. This movement initiates a wave-like motion in the plume boundary that propagates downstream from 24 to 43ms. The thermal plume appears to have two distinct cloud regions at 43ms; an upper, "buoyancy-driven" cloud and a lower, "ion-driven" cloud. The "ion-driven" cloud is the region in the thermal plume where the electrical body force has become established and is effectively driving the bulk gas downstream. The increasing of axial velocity induces an entrainment of cool gas from the ambient environment which is the cause of the reduction in the plume diameter. The plume diameter is determined by the inward radial convection of cool entrainment gas balancing with the outward radial thermal/mass diffusion of hot combustion gases. While the velocity profile is quantitatively undetermined, the profile of this "ion-driven" cloud can be seen to overtake the "buoyancy-driven" cloud around 40ms. The slower "buoyancy-driven" cloud is forced outward, creating a "winged-shaped" profile from 43 to 52ms. Around 50ms, a second wave-front forms which can be seen as a small ripple. This secondary wave is most likely due to the stabilization of the flame where the parameters influencing combustion have been altered by the change in the near-flame hydrodynamics. After the second wave reaches the downstream electrode, the thermal plume reaches a new steady state ~90ms after the electrical field was switched on. The final plume boundary diameter is ~8 mm. The time it takes this first wave to travel

	from the fuel tube to the downstream electrode is ~50 ms minus 10 ms, the latter being the instant of initial movement. The interpretation of this is the time it takes a volume of gas driven by ion wind forces to travel through the electrode gap. Recalling from the time-scale analysis section, 40 ms agrees well with the predicted 34 ms as analysis. The average velocity of the wave is then t = 3.2 cm/0.040 s = 80cm/s. A voltage step producing a negative field shows the thermal plume in its initial state before the electric field is turned on. The initial plume diameter is consistently 12mm downstream of the burner. At t=0ms, the electric field is turned on. The first noticeable movement occurs at ~9ms. The base of the plume drops lower than its initial height becoming attached to the burner. As this happens, the diameter of the plume boundary near the fuel tube widens as the diameter further downstream thins. The final plume profile is achieved at ~90ms. The lower portion of the plume is wider and rounder than the upper portion. The widest diameter of the lower portion is ~13mm while the upper portion has a diameter is ~10.5mm. For a negative field, the positive ions are pulled toward the burner while electrons are pulled downstream. Near the flame, the positive ions are forced to collide with the gas exiting from the fuel tube. This decreases the axial velocity of the bulk gas in this region, which explains the widening of the plume boundary near the flame. Further downstream, the plume boundary is thinner. An ion-wind effect is occurring in this region due to the current of electrons being collected at the downstream electrode. The ion-wind effect due to electrons is not as significant as seen with positive ion due to electron's higher mobility. At 92ms the ion-wind effect cannot be disregarded. The transient applied voltage and current measurements versus time in milliseconds are recorded during this video. An interesting thing to note in the curve is the slow decay in the ion current measurement. The ion current measured in positive field shows a very fast settling time while the ion current in negative field has a very long drawn out decay. A high speed video of the methane flame during a negative field step is analyzed to explain this decay. The video starts from the flame at its initial state before the electric field is switched on. It takes ~8ms for a noticeable change in the flame structure. The initial change shows the flame height decrease as the diameter increases. As this happens, the glowing soot region increases from 0 to 40ms. Around 40ms the flame reaches a minimum height and then reverses direction till 136ms. The flame structure continues to expand up to ~130ms where its final shape is formed. The soot region was also seen to drastically decrease between 40ms and 136ms. In fact, the soot region continues to decay very slowly even after the flame structure has reached its final shape. This slow decay ends ~350ms after the electric field was initially turned on. The final state of the flame has a larger surface area and is less luminous than the original state. Visual traces of soot have completely disappeared. It is also interesting to point out that the slow decay of the soot region corresponds extremely well with the slow decay in transient ion-current measurement versus time curve. The overall flame shape has more of a "half-egg" shape compared to the initial shape. More will be done in the coming cycle to understand this soot disappearance.
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